

Ashvath Raj Ramesh

12 Champagne Ct, Watervliet, NY 12189

518-334-3700 | ashvath129@gmail.com | [linkedin.com/in/ashvathramesh](https://www.linkedin.com/in/ashvathramesh) | github.com/aramesh129

EDUCATION

Purdue University

Bachelor of Science in Honors Computer Science and Data Science | GPA: 3.7

West Lafayette, IN

Expected Graduation: May 2028

RELEVANT COURSEWORK

Data Structures & Algorithms | LLMs & Generative AI | Programming in C | Object-Oriented Programming

EXPERIENCE

Software Engineering Intern

June 2026 – Present

General Dynamics Information Technology

Albany, NY

- Enhanced in-house RAG pipeline with hybrid hierarchical-atomic chunking to support Excel test documentation
- Built a prototype RAG system (Streamlit, FastAPI, OpenAI embeddings, Qdrant) to validate chunking strategy
- Pitched prototype results to leadership and led integration of test documentation into our production RAG model using AWS Bedrock, BGE-M3, FTS5, Qdrant, and PageIndex enabling reliable retrieval across 2,000+ test cases
- Developed Selenium automation scripts to test critical scenarios, reducing the manual quality assurance effort
- Piloted AWS Kiro, authoring steering files with code and security standards to automate scans on every change

Client Solutions Developer

Aug 2025 – Jan 2026

Purdue Small Business Engineering Club

West Lafayette, IN

- Engineered a bilingual web platform for a business client, now used to manage scheduling for 500+ clients
- Built a Google Apps Script backend with dynamic template routing and a JSON API for calendar availability
- Integrated the Google Calendar API to automate event creation for confirmed appointments on owners calendar
- Automated appointment logging and confirmation emails to clients and owner using MailApp and the Sheets API

PROJECTS

Ledgr | React, Convex, Python, Flask, Node.js, Plaid, Ollama

July 2026 – Present

- Built a personal finance platform with real time data sync via Convex and per user authorization on all queries
- Architected 3 decoupled microservices: Python PDF parser, local LLM inference service, and Plaid OAuth proxy
- Engineered a local inference pipeline using Ollama served LLaMA 3.2, enforcing structured JSON output via schema validation and retry logic to generate spending insights and trend summaries from raw transaction data
- Built a unified investment tracker normalizing gain/loss across stocks, ETFs, and crypto with live price refresh

FightIQ | Python, Scikit-learn, FastAPI, Supabase, Next.js, Vercel

June 2026 – Aug 2026

- Built an end to end UFC fight prediction engine serving ML generated win probabilities for upcoming fights
- Trained a scikit-learn voting ensemble on 8,500+ bouts with SHAP explainability and Monte Carlo simulations
- Engineered multiple predictive features using exponentially weighted rolling averages to capture recent fighter form
- Automated weekly retraining with a performance gate blocking regressions in accuracy compared to production

Cloud Portfolio | AWS, Terraform, Python, GitHub Actions

May 2026 – Present

- Deployed a personal serverless portfolio on AWS using S3, CloudFront, Lambda, API Gateway, and DynamoDB
- Provisioned AWS infrastructure with Terraform enabling reproducible and version controlled deployments
- Built a CI/CD pipeline with GitHub Actions automatically deploying all changes on every push to main branch
- Implemented a REST endpoint using Lambda and DynamoDB atomic updates to power a live visitor counter

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (2026) | AWS Certified AI Practitioner (2026) | AWS Certified Cloud Practitioner (2026) | Google AI Essentials Specialization (2026) | Google Data Analytics Certificate (2024)

SKILLS

Languages: Java, JavaScript, Python, C, C++, C#, HTML/CSS, SQL, R

Frameworks & Libraries: React, Angular, FastAPI, Pandas, scikit-learn, Streamlit, JUnit

DevOps: AWS (S3, Lambda, Bedrock, DynamoDB, CloudFront), Terraform, Docker, Vercel, GitHub Actions

Databases: PostgreSQL, Oracle SQL, MS SQL, Supabase